

MindMate: Exploring the Effect of Conversational Agents on Reflective Writing

Seyed Parsa Neshaei
EPFL
Lausanne, Switzerland
seyed.neshaei@epfl.ch

Hind El Bouchrifi
EPFL
Lausanne, Switzerland
hind.elbouchrifi@epfl.ch

Thiemo Wambsganss
Institute for Digital Technology Management, Bern
University of Applied Sciences
Bern, Switzerland
thiemo.wambsganss@bfh.ch

Tanja Käser
EPFL
Lausanne, Switzerland
tanja.kaeser@epfl.ch

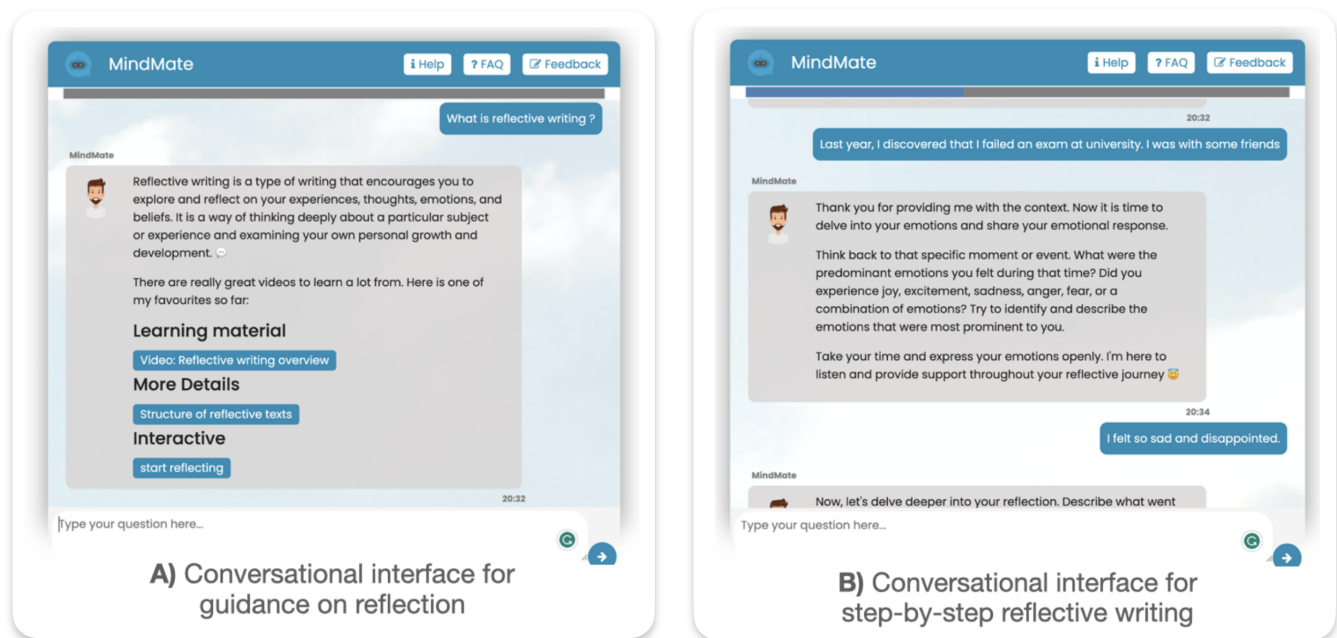


Figure 1: An overview of MindMate, our reflective writing tool, consisting of two conversational agents to help learners in the reflective writing *guidance* (A) and *writing* (B) phases.

Abstract

Reflective writing is known to be effective in improving students' learning process. However, learners often encounter difficulties when reflecting, struggling to formulate their thoughts constructively. While conversational agents have been shown to be beneficial in terms of learning gains across multiple domains, prior

works have also acknowledged downsides of using them for certain scenarios, e.g., due to cognitive overload or reduced sense of ownership. To investigate these potential effects, we designed MindMate, a conversational agent (CA) for reflective writing. We conducted a field experiment with 55 students to gain early insights on two design versions of MindMate against a static tool, by manipulating the user interface. We found a higher reflective writing quality, intention to use, and interactional enjoyment in the group *not* using a CA for the writing phase. Our results pave the way for further exploration of the effects and risks of using CAs in supporting reflective writing of learners.¹

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '25, Yokohama, Japan

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3720029>

¹The supplementary material for this paper can be found in our GitHub repository: <https://github.com/epfl-ml4ed/mindmate>

CCS Concepts

• **Human-centered computing** → **Natural language interfaces**; **Field studies**.

Keywords

conversational agents, reflective writing, intelligent writing assistants, natural language processing

ACM Reference Format:

Seyed Parsa Neshaei, Thiemo Wambsganss, Hind El Bouchrifi, and Tanja Käser. 2025. MindMate: Exploring the Effect of Conversational Agents on Reflective Writing. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 9 pages. <https://doi.org/10.1145/3706599.3720029>

1 Introduction and Related Work

Reflective writing is a process in which individuals express their thoughts, experiences, and insights on particular events [70]. It helps humans analyze past experiences to gain a deeper understanding of their skills and values, hence enhancing learning and personal growth [14]. Reflective writing is also known to improve students' metacognition [47], which is of big interest to educators and has become an important part of the learning process [49], specifically in vocational schools [11, 12, 19, 40]. With that said, writing reflective texts is not an easy endeavor, especially for novice learners. Research has shown that humans in general struggle with how to use reflective models to formulate their thoughts and emotions constructively [1, 41, 51].

Previous studies, especially in the areas of educational technology and human-computer interaction (HCI), explored different writing assistants and intelligent tutoring systems (ITS) [6, 26]. These systems utilize automated, computational approaches to help students in their writing task (e.g. reflective writing). Such systems are usually supported by machine learning (ML) models, e.g., Magooda et al. [35] used RoBERTa and DistilBERT to model the reflection quality of student texts, or Alrashidi et al. [5] provided a reflective writing rubric for computational assessment by ML models.

With that said, another solution avenue for the design and implementation of natural language processing-based ITS for reflection writing, which is the focus of this work, could be embedding a conversational agent (CA). CAs (a.k.a. *chatbots*) [37] are systems that interact with users in a dialog-based way, thus offering adaptivity and flexibility in an anthropomorphized style. CAs have been widely used in education and HCI [21, 71], such as to support language-learning [20], teach mathematics [69], and improve argumentation skills [61, 65]. Specifically, when embedded in pedagogical scenarios, CAs can emulate human instructors by providing instructions or asking clarifying questions [66]. This approach fosters personalized interactions with students, contributing to improved learning experiences and higher interactional enjoyment [61], which is also supported by theories such as the ICAP framework [13]. Also, researchers have been building up on conversational user interaction to improve reflective writing [24, 25], e.g., for apprentices [73, 75]. Wolfbauer et al. [74] found first indications that repeated use of

a CA for reflective writing positively affects the learners and increases their reflection competence. As another example, Kim et al. [24] designed a reflection CA for the mental health domain and conducted a qualitative study, finding that the CA was able to support patients in improving their daily record writing.

With that said, the literature suggests that embedding intelligent support may not always be beneficial for learners, as it may cause cognitive overload under certain conditions [18] or decrease the sense of ownership over writings [27]. They have also mentioned challenges including limitations in addressing complex questions [32] and an observed lack of humane connection [23], possibly leading users to prefer interaction with real humans [52]. Thus, there is no guarantee that embedding a CA at any given step of an ITS necessarily helps learners. Hence, deeper design studies that explore the effects of embedding CAs in each stage of reflective writing assistants are still lacking. We aim to move in the direction of tackling this research gap by answering the following research questions: **RQ1**) How does providing a CA interface for the task of reflective writing help learners write higher-quality reflective texts? **RQ2**) What are the effects of embedding reflective writing support into CA user interfaces on learners' experience² with such tools?

To find the answer to our research questions, we designed MindMate, an assistant for writing reflective texts. We used the Gibbs reflective cycle framework [16], known for its usefulness and helpfulness in a structured reflection approach [1, 3, 7, 36], in our tool (the classes of the Gibbs cycle can be seen in Table 1). We designed MindMate around two main conversational components, corresponding to two phases of writing assistants extracted from the literature: 1) a CA for the *guidance* phase [43, 44, 76], initially guiding and educating users on the significance of reflective writing and the Gibbs model, and 2) a CA for the *writing* phase [29, 35, 56], facilitating interactive creation of reflective texts through back-and-forth conversation according to the Gibbs reflective cycle.

Our single-intervention field experiment to find early insights into the effects of CA in reflective writing involved three study groups. The first group used the full MindMate (version V_1). The second group used MindMate (version V_2) with a CA for the *guidance* phase, but used traditional non-conversational text boxes for the *writing* phase. Finally, the control group (CG) did not use the CA or any intelligent support for any of the two phases, resembling the standard way and interface of how the students in our study usually wrote their reflective texts. We conducted our experiment in a vocational school with 55 nursing and caring apprentices. We asked the students to reflect on a real experience from their practical work and provided them with different interaction paradigms based on the group they belonged to. After the reflection writing, we conducted a post-survey to capture user experience. We also asked them open questions to find out what they did or did not like in our tool. Then, we analyzed the reflective writing quality of students according to the Gibbs framework.

Our results indicate that using V_2 (utilizing a CA for *guidance*, without the constraint of a back-and-forth conversational interface during the reflection *writing* phase) yielded a higher reflective text

²We use the term *experience* interchangeably with *perception* in our paper. We specifically measure *experience* in terms of a set of constructs selected from prior works, as can be seen in Figure 2.

Table 1: The description of the six components of the Gibbs reflection cycle [16].

Component	Details
Description	This section includes a presentation of the event the learner is reflecting on.
Feelings	This section includes any feelings the learners had before, at the time of, and after the event, as well as their thoughts when they were in the situation.
Evaluation	This section includes an honest opinion on the positive or negative points of the response the learner provided at the time of the event.
Analysis	This section includes possible reasons for the points mentioned in the Evaluation section. Learners may refer to references supporting the provided causes and include them in their writing in this section.
Conclusions	This section aims to summarize what happened and what the learner had gained from the event.
Action Plan	This section includes opinions on what the learner would do differently in the next time they are faced with a similar situation.

quality than the control group, according to the Gibbs framework. Participants using V_2 also reported a significantly higher intention to use and interactional enjoyment compared to those using V_1 . We also clustered participants' responses to open-ended questions, highlighting positive feedback for MindMate and their suggestions for improvement. Our early findings and insights shed light on the possible design directions and important areas of focus in future CAs for reflective writing support, and pave the way for studies exploring the benefits and risks of embedding automated feedback tools in the form of CAs in various phases of educational writing assistants.

2 MindMate: Design and Implementation of a CA for Reflective Writing

We designed MindMate around two CA interfaces, inspired by previous works [61, 65, 75]: 1) a CA in the *guidance* phase, which generally guides learners on the importance of reflective writing, reflection theory, reflection framework, and how to undertake reflection better and more efficiently (part A in Figure 1), and 2) a CA in the *writing* phase, which iteratively helps the learner to form a reflective text based on the reflection framework through back-and-forth conversation with the learner (part B in Figure 1).

User Interviews. To inform our design for MindMate, we conducted user interviews of approximately 15 minutes with ten students (3 identified as female and 7 as male, all from 19 to 24 years old). All interviewees were selected from two Western European universities and were master's or bachelor's students recruited through our contacts on campus. All of them had conducted reflection before.

To design our interview guidelines, we chose an approach similar to what Wambsgans et al. [64] conducted, and adapted it to our reflective writing problem space. We specifically asked questions on their prior experience with reflection, their motivations, what they consider as a *good* reflection, if they think reflection can improve learning, and what skill or assistance type can help them in assessing learning quality. Then, we asked them to provide their

ideas on what can make them uncomfortable with communicating experiences with an AI agent, how a chatbot might be able to help individuals undertake better reflections compared to human conversations, what characteristics they expect from the tool to provide them with a better experience, and how a tool giving tutoring on their reflection skills should look like.

This small-scale user interview process revealed that the users wished for an easy-to-use and intuitive interface for reflective writing. Moreover, they expected the tool to provide the necessary guidance and tutoring about reflection theory and how to effectively write self-reflections with clear and diverse content. Also, they wished the tool would provide learners with an interactive experience, where they are guided through their reflection process. Finally, they mentioned the necessity of providing trust-enhancing components to ensure data privacy.

Interaction Design. Based on the design interview insights, we developed MindMate as an online web application using HTML, CSS, and JavaScript for the front-end and the Python Flask framework for the back-end. When using MindMate, users first see a page including data privacy information, and afterward, use the *guidance* phase CA (part A, Figure 1) that provides general guidance on the importance of reflective writing and the Gibbs framework.

We used an open-source Python library called Chatomatic³ for developing the core of our CA. Two researchers, who were experts in the Gibbs reflection cycle framework, prepared a list of intents (out of which 16 different intents were directly related to instructing the learners on reflective writing) to be used as the data source of the Chatomatic library to be able to provide accurate replies in our CA. Using an intent-based approach, similar to how previous HCI research on CAs have incorporated it [4, 58, 61], enabled the users of MindMate to be able to actively interact with our CA by asking questions and receiving relevant answers by matching the most relevant intent in our back-end. The contents of the intents we used were inspired and guided by the reflection prompts used in the literature [50] and finalized by two researchers collaboratively.

³<https://github.com/spneshaei/Chatomatic>

This approach is different from directly using a large language model (LLM) to generate the responses, as it retrieves the responses from a database, rather than generating tokens of the responses on the fly. We did not use generative LLMs as domain-independent models, to have more control over what the system presents to the user so that no user of a minority or specific condition would be harmed, as well as to reduce the possibility of biases [34, 63, 68, 77] or hallucinations [24, 42, 45, 78]⁴.

Afterward, for V_1 , users would enter another CA which asks questions on different sections in the Gibbs framework to form the reflection step-by-step iteratively (part B, Figure 1), but in V_2 , they would enter a static page when they can write their reflective text inside traditional text boxes. Finally, at the end of the intervention for each version of MindMate, the users got access to a writing feedback dashboard delivering domain-independent objectivity and sentiment metrics. The dashboard was inspired by similar writing assistant works [30, 39, 48, 57, 60, 62] and was developed using the third-party code scripts available in the Spacy and NLTK code libraries⁵.

3 Experimental In-Class Evaluation

To find early insights into how a CA interface helps learners in the quality of their reflective writings (RQ1) and how it affects user experience and perception (RQ2), we conducted an experimental classroom field study. A summary of the experimental setup can be seen in Figure 2.

3.1 Participants

We conducted an in-person field experiment of 60 to 90 minutes (in a similar fashion to Winkler et al. [72]) on 60 students, doing their apprenticeship in the domain of nursing and caring, across three classes in a Western European vocational school. The students were randomly allocated into the three classes at the beginning of the semester. Five students did not complete the study, leading to a final count of 55 students. We asked the students to reflect on a previous experience and provided them with different kinds of interaction paradigms for reflection writing, based on their assigned experiment group (V_1 , V_2 , or CG). Table 3 in the Appendix presents the counts and demographics of the groups.

3.2 Study Design

Pre-survey. In the pre-survey, we controlled the effectiveness of the randomization using two different constructs: *information technology usage model* (e.g., “If I heard about a new information technology, I would look for ways to experiment with it”) [2], and *feedback-seeking* (e.g., “It is important for me to receive feedback on my progress and learning potential”) [8]. These two pre-survey constructs have also been used in previous works on user-centric systems [54, 59, 62] and conversational agents [61] to ensure valid randomization for their studies. Additionally, we asked questions on their perception towards challenging tasks (e.g., “I like challenging and difficult tasks where I can learn new skills”).

⁴We call for future work to integrate LLMs, e.g., the GPT family of models, into the conversational agent to see if the more natural and human-like responses of such models have any effects on the outcomes of our study.

⁵The users did not use the CA again, after receiving feedback in the dashboard. We leave exploring the possible effects of the feedback module for future work.

Reflective Writing Task. After answering the pre-survey, the participants were instructed to write a reflective writing essay about a past experience. They composed a reflective essay using either version V_1 or V_2 of our tool, or the static tool (for those in CG), focusing on a particular experience they encountered while caring for patients at their nursing center during their work.

Post-survey. The experiment ended with a post-survey measuring the users’ perception of our tool and whether they would prefer to continue using it in their educational environment. Following the previously-used perception measurements capturing the experiences of learners in the literature, we used four constructs for this paper, which are also used in previous works on user-centric systems⁶ [54, 59, 62] and conversational agents [61, 67]: 1) *intention to use* (e.g., “Assuming the reflection tool was available for a next course, I would use it again”), 2) *perceived usefulness* (e.g., “I find using the reflection tool useful for writing reflective texts”), 3) *perceived ease of use* (e.g., “I find the reflection tool to be easy to interact with”), and 4) *interactional enjoyment* (e.g., “The reflection tool is fun and enjoyable to interact with”). Finally, the post-survey also contained three open questions, capturing 1) what the participants specifically liked about MindMate, 2) how MindMate affected their reflection process, and 3) what they thought could be improved.

3.3 Reflective Writing Quality Measures

We measured the quality of the reflective texts we collected from the learners as an indirect proxy for learning outcomes⁷, in line with studies that show the correlations between reflective writing and learning gains [10, 15, 22, 38]. We specifically focused on the adherence of the writings to the employed reflection framework (Gibbs reflective cycle), guided by similar approaches in previous works [31, 33, 50, 55]. Two learning sciences researchers annotated a random sample of 15% of all the reflective writings in two rounds. The annotators identified whether each sentence belongs to which of the six classes of Gibbs cycle (see Section 1), by color coding the sentences. After finishing the first 15%, we compared their annotations and found a 92% agreement. After resolving the disagreements collaboratively in a workshop, each independently annotated half of the remaining texts. The two independent measurements assigned to each text were the following: **A) Reflection structure** (*out of 6*) as an indicator of the reflection structure and how much the reflective text adheres to the Gibbs framework, granting one point for each of the six sections of the Gibbs reflective cycle, if they were present, and **B) Reflection quality** (*out of 5*) as an indicator of the quality of the reflective texts with regards to the Gibbs reflective cycle framework, assigned to the texts directly after annotating each of them. The meanings of the quality score points, collaboratively decided by the two annotator researchers in a workshop, are described in Table 2 in the Appendix.

4 Results

We first compared the pre-survey measures, to ensure valid randomization, and did not find any difference among groups. For all items in this section, we first checked normality (Shapiro-Wilk test) and

⁶We leave exploring other constructs for future work.

⁷We leave investigating the learning outcomes directly (e.g., using a test) for future work.

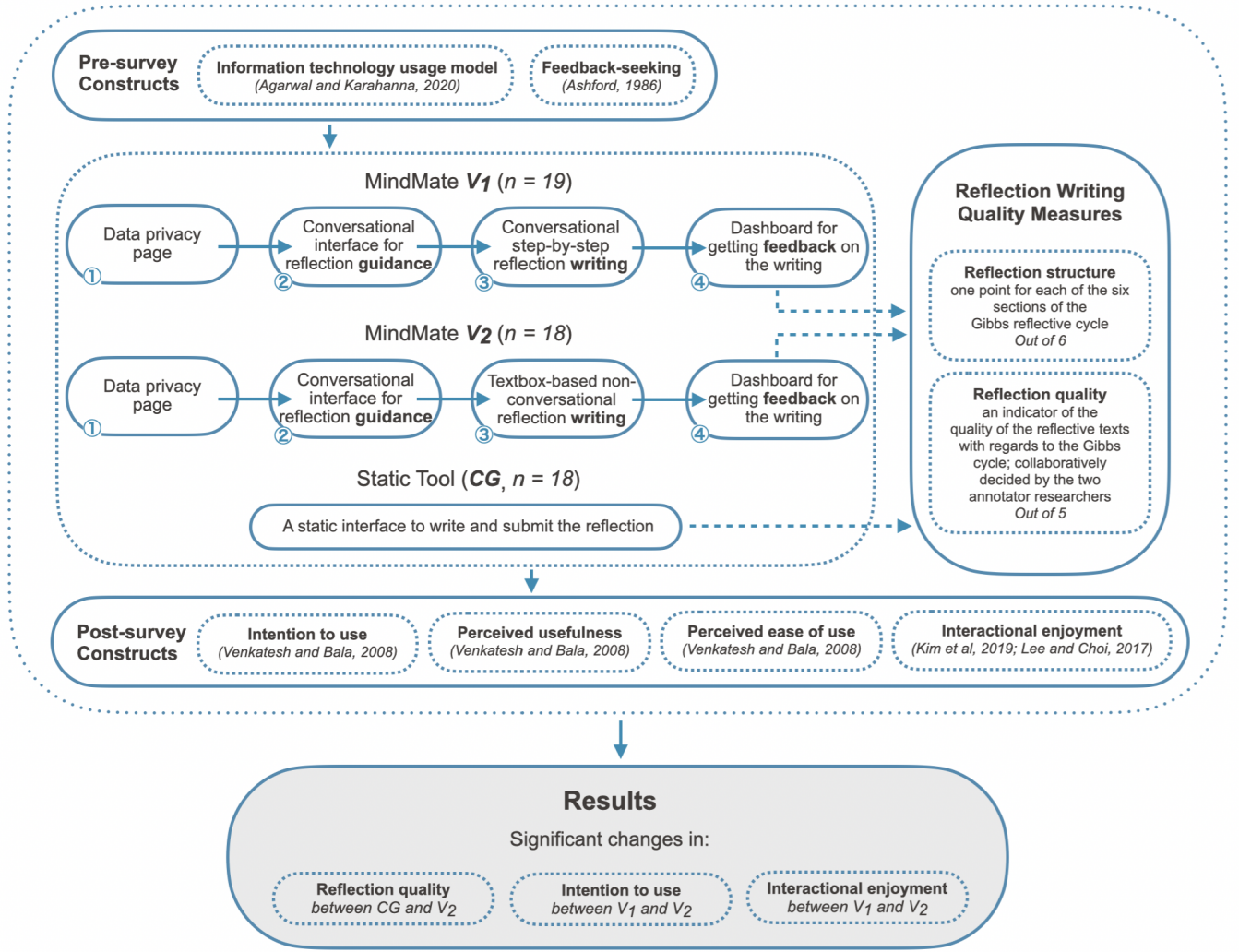


Figure 2: A summary of the experimental setup we conducted for MindMate. Participants answered the pre-survey questions, then interacted with a version of our tool based on their group, and then answered the post-survey constructs. Results indicate significant changes in the *reflection quality* between CG and V₂, as well as in two of the post-survey constructs between V₁ and V₂.

homogeneity of variance (Levene’s test). If both assumptions were met, we conducted ANOVA with post-hoc Tukey’s test. Otherwise, we used the Kruskal-Wallis test, followed by Dunn’s post-hoc test where applicable.

4.1 RQ1: Reflection Text Structure and Quality

We found that in both metrics (*reflection structure* and *reflection quality*), the mean score for those using V₂ is higher than both the other groups. We found a significant difference in terms of *reflection quality* ($p < .05$) between the three groups, and a trend to significance ($p = .08 < .10$) for *reflection structure*. The post-hoc pairwise Dunn tests revealed that the reflective writings from V₂ exhibited a significantly higher quality than CG ($p < .05$). The

full results of pairwise comparisons can be seen in Table 4 in the Appendix.

4.2 RQ2: User Perception and Experience

We found higher mean values for V₂ across all four post-survey metrics. Our analysis revealed $p < .05$ for the *intention to use* and *interactional enjoyment* metrics, and post-hoc tests showed $p < .05$ when comparing V₁ and V₂. The full results of pairwise comparisons can be seen in Table 5 in the Appendix.

4.3 Qualitative Analysis of Open Questions

We also conducted a deductive thematic qualitative analysis of the answers the participants provided to our open questions after

using MindMate in the treatment groups, and extracted three main clusters from their responses:

Ease of use and interactional enjoyment. 18.92% of the participants specifically mentioned the ease of use and the quick responses as positive aspects of working with MindMate. However, some participants (8.11%) thought MindMate was confusing and did not fully understand its functionalities. Interestingly, we found five participants in V_2 indicating they enjoyed the interaction, liked the tool, or in the case of one participant, *wouldn't change a thing about MindMate*, versus only two participants in V_1 , in line with the higher *interactional enjoyment* in V_2 .

Guiding through reflective writing. 32.43% of the participants mentioned they considered their reflective writing process with the help of MindMate as positive, indicated that MindMate made them *think* about the situation and make the reflections more precise, and also, believed that they were not *worried* about what to write next.

Less personal interaction compared to humans. While participants generally enjoyed their interaction with MindMate (based on the *interactional enjoyment* metric), some of them (10.81%) mentioned working with such a CA was superficial and that they would have preferred a personal interaction. Interestingly, all of the participants mentioning this issue belonged to V_1 , suggesting that this effect was mostly observed after introducing the CA in the *writing* phase.

5 Discussion

Drawing from the success of CAs in pedagogical settings [9, 28, 58, 61] and reflective writing [24, 25, 73], one can expect the use of CAs in our reflection scenario to enhance learner engagement and help them write higher-quality reflective texts. However, prior works have also highlighted the disadvantages of using CAs in certain pedagogical scenarios (e.g. having less humane connection or higher cognitive overload) [18, 52], rendering them less effective. Our early insights suggest that using a CA might be most effective if merely incorporated in the *guidance* phase when learners are tutored on reflection theory and the Gibbs cycle. We found that entering text in traditional non-conversational text boxes in the *writing* phase resulted in a high *reflective writing quality*, *intention to use*, and *interactional enjoyment*. We specifically examined learners' open-ended responses to justify our quantitative findings. We found that only learners who were exposed to a CA also in the *writing* phase provided comments on less humane interaction and how they found using such a system strange. Also, five participants among those using the CA only in the *guidance* phase mentioned they enjoyed interacting with the tool, versus only two among those using the CA in both phases. Thus, our early results suggest that embedding a CA in the *guidance* phase can potentially be helpful for learners; however, utilizing them for back-and-forth conversation for forming a text may be less effective than traditional text boxes. Additionally, we believe the nature of interaction with a CA plays a role in our results. The CA sequentially poses the questions, restricting the ability of the users to revisit and adjust their writings as they move forward. While acceptable for the *guidance* phase (as there is no writing task there), this limits users' control over their final text in the *writing* phase. As a result, the CA might

have undermined the users' sense of autonomy and competence when conducting the focused activity of reflective writing, which are key components of motivation based on the self-determination theory [53] and could have possibly affected the behavioral measures captured in the post-survey constructs. Previous works have acknowledged that interactions with intelligent agents may reduce the sense of ownership over texts [27] and increase cognitive overload [18], potentially explaining the open responses as well as the reduced *intention to use* and *interactional enjoyment* among those using V_1 .

Limitations. Nevertheless, our study comes with a set of limitations. Our field experiment was conducted in a single German-language school with a gender-imbalanced sample; we call for future work to also replicate our experiment in more linguistically and regionally diverse environments. We also call for future researchers to explore feedback support based on automated reflection analysis methods (e.g., [17, 46]) or to integrate generative AI models (e.g., GPT) to provide more human-like interaction in the CA, possibly changing users' perception towards the conversational interface. Finally, in future work, we aim to conduct a longitudinal study spanning an entire semester, incorporating more frequent use of MindMate to gather comprehensive insights on user behavior and learning effects.

6 Conclusion

While reflective writing is a beneficial metacognitive activity, many learners struggle to reflect effectively. CAs have shown promise in enhancing learning across domains, but risk cognitive overload and reduced sense of ownership. In a classroom study of MindMate, our reflective writing assistant, students who used a CA for *guidance* but not *writing* showed higher *reflective writing quality*, *intention to use*, and *interactional enjoyment*. Our findings highlight the impacts of CAs on reflective writing practices, emphasizing careful consideration for embedding them in the *writing* phase. Our research lays the foundation for future studies into the advantages and challenges of integrating CAs into different stages of educational writing support systems.

Acknowledgments

This project was substantially financed by the Swiss State Secretariat for Education, Research, and Innovation (SERI).

References

- [1] Ikin Syamsudin Adeani, R Bunga Febriani, and Syafryadin Syafryadin. 2020. Using GIBBS' reflective cycle in making reflections of literary analysis. *Indonesian EFL Journal* 6, 2 (2020), 139–148.
- [2] Ritu Agarwal and Elena Karahanna. 2000. Time flies when you're having fun: Cognitive absorption and beliefs about information technology usage. *MIS quarterly* (2000), 665–694. Publisher: JSTOR.
- [3] Abdelhamid M Ahmed. 2019. Students' reflective journaling: An impactful strategy that informs instructional practices in an EFL writing university context in Qatar. *Reflective Practice* 20, 4 (2019), 483–500. Publisher: Taylor & Francis.
- [4] Haritha Akkineni, PVS Lakshmi, and Lasya Sarada. 2022. Design and development of retrieval-based chatbot using sentence similarity. In *IoT and analytics for sensor networks: Proceedings of ICWSNUCA 2021*. Springer, 477–487.
- [5] Huda Alrashidi, Nouf Almujally, Methaq Kadhun, Thomas Daniel Ullmann, and Mike Joy. 2022. Evaluating an automated analysis using machine learning and natural language processing approaches to classify computer science students' reflective writing. In *Pervasive computing and social networking: Proceedings of ICPCSN 2022*. Springer, 463–477.

- [6] John R Anderson, C Franklin Boyle, and Brian J Reiser. 1985. Intelligent tutoring systems. *Science (New York, N.Y.)* 228, 4698 (1985), 456–462. Publisher: American Association for the Advancement of Science.
- [7] Siti Noor Aneis Hashim, Aizan Yaacob, Ina Suryani, Ratnawati Mohd Asraf, Zolkefli Bahador, and Nadya Supian. 2023. Exploring the use of gibbs' reflective model in enhancing in-service ESL teachers' reflective writing. *Arab World English Journal (AWEJ) Volume 14* (2023).
- [8] Susan J Ashford. 1986. Feedback-seeking in individual adaptation: A resource perspective. *Academy of Management journal* 29, 3 (1986), 465–487. Publisher: Academy of Management Briarcliff Manor, NY 10510.
- [9] Daniel Bailey. 2019. Chatbots as conversational agents in the context of language learning. *Proceedings of the fourth industrial revolution and education* (2019), 32–41.
- [10] Kirsten Berthold, Matthias Nückles, and Alexander Renkl. 2007. Do learning protocols support learning strategies and outcomes? The role of cognitive and metacognitive prompts. *Learning and Instruction* 17, 5 (2007), 564–577. Publisher: Elsevier.
- [11] Alberto Cattaneo and Elena Boldrini. 2016. Individual and collaborative writing-to-learn activities in vocational education: An overview of different instructional strategies. *Writing for professional development* (2016), 188–208. Publisher: Brill.
- [12] Alberto AP Cattaneo and Elisa Motta. 2021. "I reflect, therefore I am... a good professional". On the relationship between reflection-on-action, reflection-in-action and professional performance in vocational education. *Vocations and Learning* 14, 2 (2021), 185–204. Publisher: Springer.
- [13] Michelene TH Chi and Ruth Wylie. 2014. The ICAP framework: Linking cognitive engagement to active learning outcomes. *Educational psychologist* 49, 4 (2014), 219–243. Publisher: Taylor & Francis.
- [14] Jordi Colomer, Teresa Serra, Dolors Cañabate, and Remigijus Bubnys. 2020. Reflective learning in higher education: Active methodologies for transformative practices. *Sustainability* (2020). Issue: 9 Pages: 3827 Volume: 12.
- [15] Kristie R Dukewich and Deborah P Vossen. 2015. Toward accuracy, depth and insight: how reflective writing assignments can be used to address multiple learning objectives in small and large courses. *Collected Essays on Learning and Teaching* 8 (2015), 97–110.
- [16] Graham Gibbs. 1988. Learning by doing: A guide to teaching and learning methods. *Further Education Unit* (1988). Publisher: Oxford Polytechnic.
- [17] Andrew Gibson, Lance De Vine, Miguel Canizares, and Jill Willis. 2023. Reflexive expressions: Towards the analysis of reflexive capability from reflective text. In *International conference on artificial intelligence in education*. Springer, 353–364.
- [18] Songhee Han and Min Kyung Lee. 2022. FAQ chatbot and inclusive learning in massive open online courses. *Computers & Education* 179 (2022), 104395. Publisher: Elsevier.
- [19] Mandy Hommel, Bärbel Fürstenau, and Regina H Mulder. 2023. Reflection at work—A conceptual model and the meaning of its components in the domain of VET teachers. *Frontiers in Psychology* 13 (2023), 923888. Publisher: Frontiers.
- [20] Weijiao Huang, Khe Foon Hew, and Luke K Fryer. 2022. Chatbots for language learning—Are they really useful? A systematic review of chatbot-supported language learning. *Journal of Computer Assisted Learning* 38, 1 (2022), 237–257. Publisher: Wiley Online Library.
- [21] Gwo-Jen Hwang and Ching-Yi Chang. 2021. A review of opportunities and challenges of chatbots in education. *Interactive Learning Environments* (2021), 1–14. Publisher: Taylor & Francis.
- [22] Sandra Hübner, Matthias Nückles, and Alexander Renkl. 2010. Writing learning journals: Instructional support to overcome learning-strategy deficits. *Learning and Instruction* 20, 1 (2010), 18–29. Publisher: Elsevier.
- [23] Augustine Osamor Ifechebuegu, Peace Kulume, and Perpetua Cherukut. 2023. Chatbots and AI in Education (AIED) tools: The good, the bad, and the ugly. *Journal of Applied Learning and Teaching* 6, 2 (2023).
- [24] Taewan Kim, Seolyeong Bae, Hyun Ah Kim, Su-woo Lee, Hwajung Hong, Chanmo Yang, and Young-Ho Kim. 2024. MindfulDiary: Harnessing large language model to support psychiatric patients' journaling. In *Proceedings of the CHI conference on human factors in computing systems*. 1–20.
- [25] Rafal Kocielnik, Lillian Xiao, Daniel Avrahami, and Gary Hsieh. 2018. Reflection companion: a conversational system for engaging users in reflection on physical activity. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 2, 2 (2018), 1–26. Publisher: ACM New York, NY, USA.
- [26] Mina Lee, Katy Ilonka Gero, John Joon Young Chung, Simon Buckingham Shum, Vipul Raheja, Hua Shen, Subhashini Venugopalan, Thiemo Wambsgans, David Zhou, Emad A Alghamdi, and others. 2024. A design space for intelligent and interactive writing assistants. In *Proceedings of the CHI conference on human factors in computing systems*. 1–35.
- [27] Mina Lee, Percy Liang, and Qian Yang. 2022. CoAuthor: Designing a human-AI collaborative writing dataset for exploring language model capabilities. In *Proceedings of the 2022 CHI conference on human factors in computing systems*. 1–19.
- [28] Haiying Li and Arthur C Graesser. 2021. The impact of conversational agents' language on summary writing. *Journal of Research on Technology in Education* 53, 1 (2021), 44–66. Publisher: Taylor & Francis.
- [29] Chenchen Liu, Jierui Hou, Yun-Fang Tu, Youmei Wang, and Gwo-Jen Hwang. 2021. Incorporating a reflective thinking promoting mechanism into artificial intelligence-supported English writing environments. *Interactive Learning Environments* (2021), 1–19. Publisher: Taylor & Francis.
- [30] Chengzhong Liu, Shixu Zhou, Dingdong Liu, Junze Li, Zeyu Huang, and Xiaojuan Ma. 2023. CoArgue: Fostering lurkers' contribution to collective arguments in community-based QA platforms. In *Proceedings of the 2023 CHI conference on human factors in computing systems*. 1–17.
- [31] Ming Liu, Simon Buckingham Shum, Efi Mantzourani, and Cherie Lucas. 2019. Evaluating machine learning approaches to classify pharmacy students' reflective statements. In *Artificial Intelligence in Education: 20th international conference, AIED 2019, Chicago, IL, USA, June 25-29, 2019, Proceedings, Part I* 20. Springer, 220–230.
- [32] Terrance Lopez and Meer Qamber. 2022. The benefits and drawbacks of implementing chatbots in higher education: A case study for international students at Jönköping University.
- [33] Cherie Lucas, Sinthia Bosnic-Anticevich, Carl R Schneider, Kathryn Bartimote-Aufflick, Mark McEntee, and Lorraine Smith. 2017. Inter-rater reliability of a reflective rubric to assess pharmacy students' reflective thinking. *Currents in Pharmacy Teaching and Learning* 9, 6 (2017), 989–995. Publisher: Elsevier.
- [34] Li Lucy and David Bamman. 2021. Gender and representation bias in GPT-3 generated stories. In *Proceedings of the third workshop on narrative understanding*. 48–55.
- [35] Ahmed Magooda, Diane Litman, Ahmed Ashraf, and Muhsin Menekse. 2022. Improving the quality of students' written reflections using natural language processing: model design and classroom evaluation. In *International conference on artificial intelligence in education*. Springer, 519–525.
- [36] Pihla Markkanen, Maritta Välimäki, Minna Anttila, and Marko Kuuskorpi. 2020. A reflective cycle: Understanding challenging situations in a school setting. *Educational Research* 62, 1 (2020), 46–62. Publisher: Taylor & Francis.
- [37] Dominic W Massaro, Michael M Cohen, Sharon Daniel, and Ronald A Cole. 1999. Developing and evaluating conversational agents. In *Human performance and ergonomics*. Elsevier, 173–194.
- [38] Pamela A McKinney and Barbara A Sen. 2012. Reflection for learning: understanding the value of reflective writing for information literacy development. *Journal of Information Literacy* 6, 2 (2012), 110–129. Publisher: Chartered Institute of Library and Information Professionals.
- [39] Paola Mejia-Domenzain, Jibril Frej, Seyed Parsa Neshaei, Luca Mouchel, Tanya Nazaretsky, Thiemo Wambsgans, Antoine Bosselut, and Tanja Käser. 2024. Enhancing procedural writing through personalized example retrieval: a case study on cooking recipes. *International Journal of Artificial Intelligence in Education* (2024), 1–37. Publisher: Springer.
- [40] Paola Mejia-Domenzain, Mirko Marras, Christian Giang, Alberto Cattaneo, and Tanja Käser. 2022. Evolutionary clustering of apprentices' self-regulated learning behavior in learning journals. *IEEE Transactions on Learning Technologies* 15, 5 (2022), 579–593. Publisher: IEEE.
- [41] Rebekkah Middleton. 2017. Critical reflection: the struggle of a practice developer. *International Practice Development Journal* (2017).
- [42] Sewon Min, Kalpesh Krishna, Xinxu Lyu, Mike Lewis, Wen-tau Yih, Pang Wei Koh, Mohit Iyyer, Luke Zettlemoyer, and Hannaneh Hajishirzi. 2023. FactScore: Fine-grained atomic evaluation of factual precision in long form text generation. *arXiv preprint arXiv:2305.14251* (2023).
- [43] Anita D Misra-Hebert, J Harry Isaacson, Martin Kohn, Alan L Hull, Mohammadreza Hojat, Klara K Papp, and Leonard Calabrese. 2012. Improving empathy of physicians through guided reflective writing. *Department of Psychiatry and Human Behavior Faculty Papers. Paper 51* (2012). <https://jdc.jefferson.edu/phbfp/51>
- [44] Jase Moussa-Inaty. 2015. Reflective writing through the use of guiding questions. *International Journal of Teaching and Learning in Higher Education* 27, 1 (2015), 104–113. Publisher: ERIC.
- [45] Dor Muhlgay, Ori Ram, Inbal Magar, Yoav Levine, Nir Ratner, Yonatan Belinkov, Omri Abend, Kevin Leyton-Brown, Amnon Shashua, and Yoav Shoham. 2024. Generating benchmarks for factuality evaluation of language models. In *Proceedings of the 18th conference of the european chapter of the association for computational linguistics (volume 1: Long papers)*, Yvette Graham and Matthew Purver (Eds.). Association for Computational Linguistics, St. Julian's, Malta, 49–66. <https://aclanthology.org/2024.eacl-long.4/>
- [46] Jan Nehyba and Michal Štefánik. 2023. Applications of deep language models for reflective writings. *Education and Information Technologies* 28, 3 (2023), 2961–2999. Publisher: Springer.
- [47] Valerie Dean O'Loughlin and Lauren Miller Griffith. 2020. Developing student metacognition through reflective writing in an upper level undergraduate anatomy course. *Anatomical Sciences Education* 13, 6 (2020), 680–693. Publisher: Wiley Online Library.
- [48] Zhenhui Peng, Qingyu Guo, Ka Wing Tsang, and Xiaojuan Ma. 2020. Exploring the effects of technological writing assistance for support providers in online mental health community. In *Proceedings of the 2020 CHI conference on human factors in computing systems*. 1–15.

- [49] John Perry, David Lundie, and Gill Golder. 2019. Metacognition in schools: what does the literature suggest about the effectiveness of teaching metacognition in schools? *Educational Review* 71, 4 (2019), 483–500. Publisher: Taylor & Francis.
- [50] Martin Pieper, Julian Roelle, Rudolf vom Hofe, Alexander Salle, and Kirsten Berthold. 2021. Feedback in reflective journals fosters reflection skills of student teachers. *Psychology Learning & Teaching* 20, 1 (2021), 107–127. Publisher: SAGE Publications Sage UK: London, England.
- [51] Julia Prior, Samuel Ferguson, and John Leaney. 2016. Reflection is hard: teaching and learning reflective practice in a software studio. In *Proceedings of the Australasian Computer Science Week Multiconference*. 1–8.
- [52] Rajat Roy and Vik Naidoo. 2021. Enhancing chatbot effectiveness: The role of anthropomorphic conversational styles and time orientation. *Journal of Business Research* 126 (2021), 23–34. Publisher: Elsevier.
- [53] Richard M Ryan and Edward L Deci. 2000. Self-determination theory and the facilitation of intrinsic motivation, social development, and well-being. *American psychologist* 55, 1 (2000), 68. Publisher: American Psychological Association.
- [54] Xiaotian Su, Thiemo Wambsganss, Roman Rietsche, Seyed Parsa Neshaei, and Tanja Käser. 2023. Reviewer: AI-Generated instructions for peer review writing. In *Proceedings of the 18th workshop on innovative use of NLP for building educational applications (BEA 2023)*. 57–71.
- [55] Cherie Tsingos, Sinthia Bosnic-Anticevich, John M Lonie, and Lorraine Smith. 2015. A model for assessing reflective practices in pharmacy education. *American journal of pharmaceutical education* 79, 8 (2015). Publisher: American Journal of Pharmaceutical Education.
- [56] Thomas Daniel Ullmann. 2019. Automated analysis of reflection in writing: Validating machine learning approaches. *International Journal of Artificial Intelligence in Education* 29, 2 (2019), 217–257. Publisher: Springer.
- [57] Thiemo Wambsganss, Andrew Caines, and Paula Buttery. 2022. ALEN app: argumentative writing support to foster English language learning. In *Proceedings of the 17th workshop on innovative use of NLP for building educational applications (BEA 2022)*. 134–140.
- [58] Thiemo Wambsganss, Sebastian Guggisberg, and Matthias Söllner. 2021. Arguebot: A conversational agent for adaptive argumentation feedback. In *Innovation through information systems: Volume II: A collection of latest research on technology issues*. Springer, 267–282.
- [59] Thiemo Wambsganss, Andreas Janson, Tanja Käser, and Jan Marco Leimeister. 2022. Improving students argumentation learning with adaptive self-evaluation nudging. *Proceedings of the ACM on Human-Computer Interaction* 6, CSCW2 (2022), 1–31. Publisher: ACM New York, NY, USA.
- [60] Thiemo Wambsganss, Andreas Janson, and Jan Marco Leimeister. 2022. Enhancing argumentative writing with automated feedback and social comparison nudging. *Computers & Education* 191 (2022), 104644. Publisher: Elsevier.
- [61] Thiemo Wambsganss, Tobias Kueng, Matthias Soellner, and Jan Marco Leimeister. 2021. ArgueTutor: An adaptive dialog-based learning system for argumentation skills. In *Proceedings of the 2021 CHI conference on human factors in computing systems*. 1–13.
- [62] Thiemo Wambsganss, Christina Niklaus, Matthias Cetto, Matthias Söllner, Siegfried Handschuh, and Jan Marco Leimeister. 2020. AL: An adaptive learning support system for argumentation skills. In *Proceedings of the 2020 CHI conference on human factors in computing systems*. 1–14.
- [63] Thiemo Wambsganss, Xiaotian Su, Vinitra Swamy, Seyed Parsa Neshaei, Roman Rietsche, and Tanja Käser. 2023. Unraveling downstream gender bias from large language models: A study on AI educational writing assistance. *arXiv preprint arXiv:2311.03311* (2023).
- [64] Thiemo Wambsganss, Matthias Söllner, Kenneth R Koedinger, and Jan Marco Leimeister. 2022. Adaptive empathy learning support in peer review scenarios. In *Proceedings of the 2022 CHI conference on human factors in computing systems*. 1–17.
- [65] Thiemo Wambsganss, Matthias Söllner, and Jan Marco Leimeister. 2020. Design and evaluation of an adaptive dialog-based tutoring system for argumentation skills. In *International conference on information systems (ICIS)-Hyderabad, india*.
- [66] Thiemo Wambsganss, Rainer Winkler, Matthias Söllner, and Jan Marco Leimeister. 2020. A conversational agent to improve response quality in course evaluations. In *Extended Abstracts of the 2020 CHI conference on human factors in computing systems*. 1–9.
- [67] Thiemo Wambsganss, Naim Zierau, Matthias Söllner, Tanja Käser, Kenneth R Koedinger, and Jan Marco Leimeister. 2022. Designing conversational evaluation tools: A comparison of text and voice modalities to improve response quality in course evaluations. *Proceedings of the ACM on Human-Computer Interaction* 6, CSCW2 (2022), 1–27. Publisher: ACM New York, NY, USA.
- [68] Yixin Wan, George Pu, Jiao Sun, Aparna Garimella, Kai-Wei Chang, and Nanyun Peng. 2023. "Kelly is a Warm Person, Joseph is a Role Model": Gender biases in LLM-Generated reference letters. *arXiv preprint arXiv:2310.09219* (2023).
- [69] Yousef Wardat, Mohammad A Tashtoush, Rommel AlAli, and Adeeb M Jarrah. 2023. ChatGPT: A revolutionary tool for teaching and learning mathematics. *Eurasia Journal of Mathematics, Science and Technology Education* 19, 7 (2023), em2286. Publisher: Modestum.
- [70] Kate Williams, Mary Woolliams, and Jane Spiro. 2020. *Reflective writing*. Bloomsbury Publishing.
- [71] Rainer Winkler and Matthias Söllner. 2018. Unleashing the potential of chatbots in education: A state-of-the-art analysis. In *Academy of management proceedings*, Vol. 2018. Academy of Management Briarcliff Manor, NY 10510, 15903. Issue: 1.
- [72] Rainer Winkler, Matthias Söllner, and Jan Marco Leimeister. 2021. Enhancing problem-solving skills with smart personal assistant technology. *Computers & Education* 165 (2021), 104148. Publisher: Elsevier.
- [73] Irmtraud Wolfbauer, Mia Magdalena Bangerl, Katharina Maitz, and Viktoria Pammer-Schindler. 2023. Rebo at work: Reflecting on working, learning, and learning goals with the reflection guidance chatbot for apprentices. In *Extended abstracts of the 2023 CHI conference on human factors in computing systems*. 1–7.
- [74] Irmtraud Wolfbauer, Viktoria Pammer-Schindler, Katharina Maitz, and Carolyn P Rosé. 2022. A script for conversational reflection guidance: A field study on developing reflection competence with apprentices. *IEEE Transactions on Learning Technologies* 15, 5 (2022), 554–566. Publisher: IEEE.
- [75] Irmtraud Wolfbauer, Viktoria Pammer-Schindler, and C Rosé. 2020. Rebo junior: analysis of dialogue structure quality for a reflection guidance chatbot. In *EC-TEL impact paper proceedings 2020: 15th European conference on technology enhanced learning*.
- [76] Yeou Min Wong, Rosnidar Mansor, and Syakirah Samsudin. 2016. The use of critical reflection manual in writing reflective journal: A case study of Malaysian student teachers' perceptions. *Geografia* 12, 1 (2016). Publisher: Universiti Kebangsaan Malaysia.
- [77] Travis Zack, Eric Lehman, Mirac Suzgun, Jorge A Rodriguez, Leo Anthony Celi, Judy Gichoya, Dan Jurafsky, Peter Szolovits, David W Bates, Raja-Elie E Abdunour, and others. 2023. Coding inequity: Assessing GPT-4's potential for perpetuating racial and gender biases in healthcare. *medRxiv : the preprint server for health sciences* (2023), 2023–07. Publisher: Cold Spring Harbor Laboratory Press.
- [78] Yue Zhang, Yafu Li, Leyang Cui, Deng Cai, Lemao Liu, Tingchen Fu, Xinting Huang, Enbo Zhao, Yu Zhang, Yulong Chen, and others. 2023. Siren's song in the AI ocean: A survey on hallucination in large language models. *arXiv preprint arXiv:2309.01219* (2023).

Appendix

Table 2: The meaning of each of the score points for the reflection quality measure according to the Gibbs model.

Score	Meaning	Description
1	Not well reflected	This score is granted when there are less than three out of six elements the text has complied with.
2	Almost not well reflected	This score is granted when there are at least four out of six elements present, but there is no conclusion or action plan.
3	Medium	This score is granted when there are at least four out of six elements present, as well as conclusion or action plan.
4	Almost well reflected	This score is granted when there are at least four out of six elements present, as well as conclusion, action plan, detailed description, and detailed analysis.
5	Well reflected	This score is granted when all elements out of six are present in the reflective text and there is a strong focus on the action plan.

Table 3: The demographics of the groups in our experimental evaluation. The *Did Not Finish* column indicates participants who did not complete the study correctly, failing to answer all questions or submit their reflective text.

Group	Tool Used	Demographics	Average Age	SD Age	Did Not Finish
V1	MindMate V ₁	17 females, 3 males	26.50	9.57	1
V2	MindMate V ₂	19 females, 1 male	21.90	4.81	2
CG	Static tool	17 females, 2 males, 1 non-binary	22.45	3.24	2

Table 4: The mean, standard deviations, and the results of conducting post-hoc tests on the reflective text quality measures. p = p-value; d = Cohen's d .

Metric / Construct	CG		V1		V2		p-value and effect size		
	Mean	SD	Mean	SD	Mean	SD	CG vs V1	CG vs V2	V1 vs V2
Reflection structure (1-6)	3.83	1.50	3.95	1.75	4.89	0.83	$p = .59$ $d = .07$	$p = .09$ $d = .87$	$p = .14$ $d = .69$
Reflection quality (1-5)	2.61	1.24	2.79	1.18	3.50	0.71	$p = .51$ $d = .15$	$p < .05$ $d = .88$	$p = .09$ $d = .73$

Table 5: The mean, standard deviations, and the results of conducting statistical post-hoc tests on the post-survey constructs. For the *perceived usefulness* and *perceived ease of use* constructs, the Kruskal-Wallis tests did not identify any significant difference, so no results of post-hoc tests or effect sizes were calculated. p = p-value; d = Cohen's d .

Metric / Construct	CG		V1		V2		p-value and effect size		
	Mean	SD	Mean	SD	Mean	SD	CG vs V1	CG vs V2	V1 vs V2
Intention to use (1-7)	4.61	1.14	3.93	1.47	5.09	0.91	$p = .23$ $d = .52$	$p = .23$ $d = .46$	$p < .05$ $d = .95$
Perceived usefulness (1-7)	4.69	1.28	3.84	1.54	4.94	0.76	-	-	-
Perceived ease of use (1-7)	4.67	1.23	4.95	1.20	5.15	0.62	-	-	-
Interactional enjoyment (1-7)	4.08	1.15	3.91	1.34	4.79	0.70	$p = .88$ $d = .14$	$p = .14$ $d = .74$	$p < .05$ $d = .82$